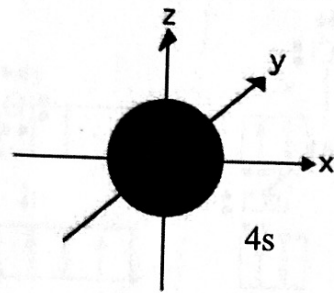
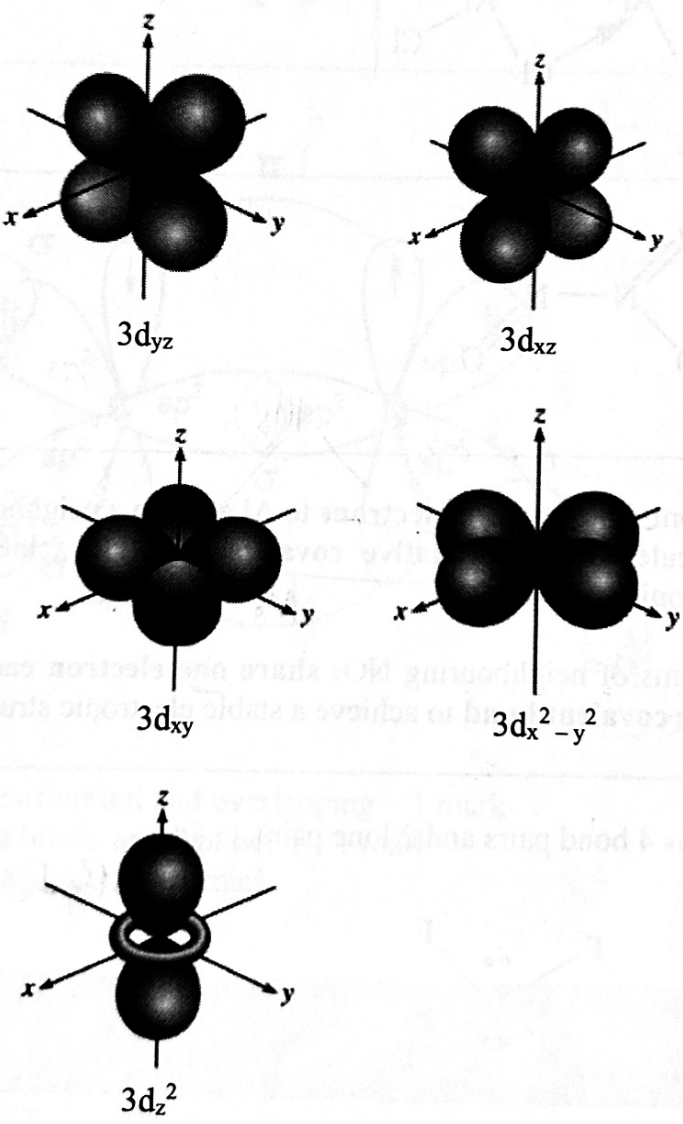


| NO. | PART | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | (a)  | <p> <i>meriti tulis</i><br/>           (Average atomic mass) = <math>\frac{\sum Q_i M_i}{\sum Q_i}</math> @ <i>if formula shj xda markah</i><br/> <math display="block">= \frac{(75.77 \times 34.969) + (24.23 \times 36.966)}{100}</math> <math display="block">= 35.45 \text{ amu}</math> <i>insist</i><br/>           (Relative atomic mass) = <math>\frac{35.45 \text{ amu}}{\frac{1}{12} \times 12.00 \text{ amu}}</math> <i>RAM = <math>\frac{\sum Q_i A_i}{\sum Q_i}</math></i><br/> <math display="block">= 35.45</math> </p> |
|     | (b)  | <p> <math>\text{BaO} + \text{H}_2\text{O} \rightarrow \text{Ba(OH)}_2</math><br/>           Mole of BaO = <math>0.0600 \text{ mol L}^{-1} \times 2.50 \text{ L}</math> @<br/> <math>= 0.150 \text{ mol}</math> <i>if sdah 2 still r kag lila</i><br/>           Mass of BaO = <math>0.150 \text{ mol} \times 153.3 \text{ g mol}^{-1}</math> <i>ecf ganti</i><br/> <math>= 23.0 \text{ g}</math> <i>(unit insit)</i> </p>                                                                                                             |
|     | (c)  | <p>           Mass of C in <math>\text{CO}_2 = \frac{12}{44} \times 207.8 \text{ g}</math><br/> <math>= 56.67 \text{ g}</math><br/>           Mass of H in <math>\text{H}_2\text{O} = \frac{2}{18} \times 85.04 \text{ g}</math><br/> <math>= 9.449 \text{ g}</math><br/>           Mass of O = <math>141.72 - (56.67 + 9.45)</math><br/> <math>= 75.60 \text{ g}</math> </p>                                                                                                                                                         |

| NO.            | PART                            | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                 |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
|----------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---|---|---|----------|-------|-------|-------|------------|---------------------------------|--------------------------------|---------------------------------|----------------|--------------------------------|--------------------------------|--------------------------------|
|                |                                 | <table><tr><td></td><td>C</td><td>H</td><td>O</td></tr><tr><td>Mass (g)</td><td>56.67</td><td>9.449</td><td>75.60</td></tr><tr><td>Mole (mol)</td><td><math>\frac{56.67}{12}</math><br/><math>= 4.723</math></td><td><math>\frac{9.449}{1}</math><br/><math>= 9.449</math></td><td><math>\frac{75.60}{16}</math><br/><math>= 4.725</math></td></tr><tr><td>Simplest ratio</td><td><math>\frac{4.723}{4.723}</math><br/><math>= 1</math></td><td><math>\frac{9.449}{4.723}</math><br/><math>= 2</math></td><td><math>\frac{4.725}{4.723}</math><br/><math>= 1</math></td></tr></table> <p>Empirical formula = <b>CH<sub>2</sub>O</b></p> <p>The empirical mass of the compounds is 30 g</p> $n = \frac{120}{30} = 4$ <p>Molecular formula = <b>C<sub>4</sub>H<sub>8</sub>O<sub>4</sub></b></p> |                                 | C | H | O | Mass (g) | 56.67 | 9.449 | 75.60 | Mole (mol) | $\frac{56.67}{12}$<br>$= 4.723$ | $\frac{9.449}{1}$<br>$= 9.449$ | $\frac{75.60}{16}$<br>$= 4.725$ | Simplest ratio | $\frac{4.723}{4.723}$<br>$= 1$ | $\frac{9.449}{4.723}$<br>$= 2$ | $\frac{4.725}{4.723}$<br>$= 1$ |
|                | C                               | H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | O                               |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
| Mass (g)       | 56.67                           | 9.449                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 75.60                           |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
| Mole (mol)     | $\frac{56.67}{12}$<br>$= 4.723$ | $\frac{9.449}{1}$<br>$= 9.449$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\frac{75.60}{16}$<br>$= 4.725$ |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
| Simplest ratio | $\frac{4.723}{4.723}$<br>$= 1$  | $\frac{9.449}{4.723}$<br>$= 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\frac{4.725}{4.723}$<br>$= 1$  |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
|                | (d)(i)                          | <p>Oxidation : <math>(\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-) \times 6</math></p> <p>Reduction : <math>\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6\text{e}^- \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O}</math></p> <p>Overall : <math>\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6 \text{Fe}^{2+} \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O} + 6 \text{Fe}^{3+}</math></p>                                                                                                                                                                                                                                                                                                                                                                             |                                 |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |
|                | (d)(ii)                         | $\frac{M_a V_a}{a} = \frac{M_b V_b}{b} \quad @$ $\frac{0.050 \times 25.0}{6} = \frac{0.0069 \times V_b}{1}$ <p>= 30 mL</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                 |   |   |   |          |       |       |       |            |                                 |                                |                                 |                |                                |                                |                                |

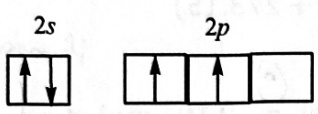
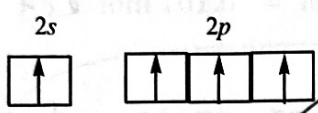
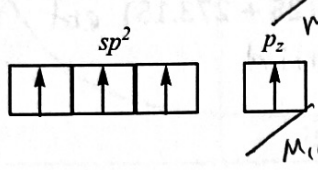
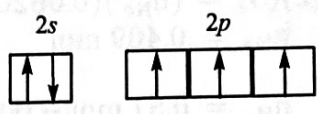
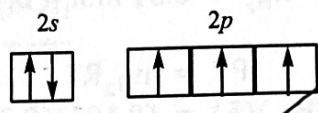
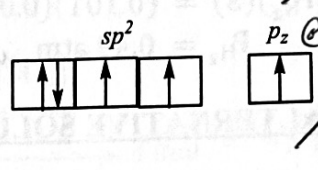
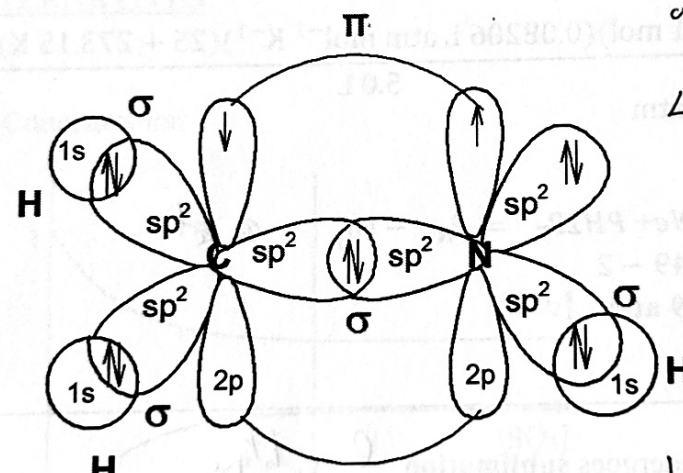
TO

| NO. | PART   | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | (a)    | $v = \frac{c}{\lambda} \quad @$ $2.74 \times 10^{14} = \frac{3.0 \times 10^8}{\lambda} \quad @$ $\lambda = 1.095 \times 10^{-6} \text{ m} \quad (\text{unit not insert})$ $\frac{1}{\lambda} = R_H \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ where } (n_1 < n_2 \text{ \& } n_1 = 3) \text{ (Paschen series)} \quad @$ $\frac{1}{1.095 \times 10^{-6}} = 1.097 \times 10^7 \left( \frac{1}{3^2} - \frac{1}{n_2^2} \right)$ $n_2 = 6$ <p>Therefore, the electron transition is from <math>n = 6</math> to <math>n = 3</math></p> <p><b><u>ALTERNATIVE SOLUTION:</u></b></p> $\Delta E = h\nu \quad @$ $= (6.6256 \times 10^{-34})(2.74 \times 10^{14} \text{ s}^{-1}) \quad @$ $= 1.815 \times 10^{-19} \text{ J}$ $\Delta E = R_H \left( \frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \text{ \& } n_f = 3 \text{ (Paschen series)} \quad @$ $-1.815 \times 10^{-19} = 2.18 \times 10^{-18} \left( \frac{1}{n_i^2} - \frac{1}{3^2} \right)$ $n_i = 6$ <p>Therefore, the electron transition is from <math>n = 6</math> to <math>n = 3</math></p> |
|     | (b)(i) | $Q^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 \quad @ [Ar]3d^6$ $Q^{3+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 \quad @ [Ar]3d^5$ <p><b><math>Q^{3+}</math> is more stable</b></p> <p><b><math>Q^{3+}</math> has half-filled <u>d</u> orbital whereas <math>Q^{2+}</math> has partially filled <u>d</u> orbital.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| NO. | PART    | SCHEME                                                                                                                                                                                                                                                                                                                                                                               |
|-----|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | (b)(ii) | <div data-bbox="502 145 1436 504"> <p>Q</p>  <p>label axis<br/>shape</p> </div> <div data-bbox="502 526 1436 1881"> <p>Q<sup>3+</sup></p>  <p><i>*Either ONE of the d-orbital shapes IS ACCEPTED</i></p> </div> |
|     |         | TOTAL                                                                                                                                                                                                                                                                                                                                                                                |



| NO. | PART     | SCHEME                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | (a)(i)   |                                                                                                                                                                                                                                                                                                                                                                                         |
|     | (a)(ii)  | <p><i>Cl-Al</i><br/><i>lone pair not in 179</i></p>                                                                                                                                                                                                                                                                                                                                     |
|     | (a)(iii) | <p><i>e- pair / MS</i></p>                                                                                                                                                                                                                                                                                                                                                              |
|     | (a)(iv)  | <p>Cl atom <u>donates both electrons</u> to Al atom in a neighbouring <math>\text{AlCl}_3</math> molecule to form a <u>dative covalent bond</u> to achieve a stable electronic structure. <i>MS any name</i></p> <p>N atoms of neighbouring <math>\text{NO}_2</math> <u>share one electron</u> each to form a single <u>covalent bond</u> to achieve a stable electronic structure.</p> |
|     | (b)      | <p>Xe has 4 bond pairs and 2 lone pairs @</p> <p>Square planar</p> <p>Bond angle in equatorial plane <math>90^\circ</math></p> <p><i><math>\text{AB}_4\text{E}_2</math> X remark.</i></p>                                                                                                                                                                                               |

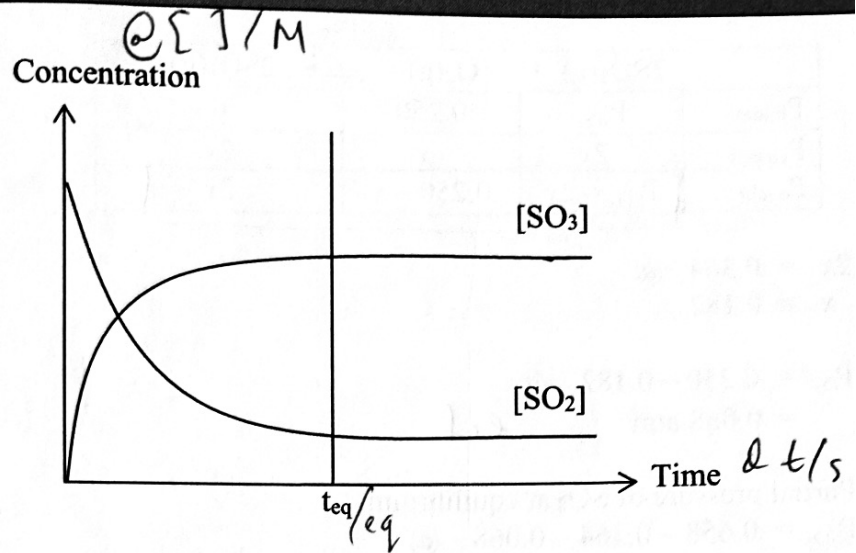
| NO.            | PART    | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MA |
|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|                | (c)(i)  | <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>Carbon</p> <p>Ground state</p>  <p>Excited state</p>  <p>Hybrid</p>  </div> <div style="text-align: center;"> <p>Nitrogen</p> <p>Ground state</p>  <p>Excited state</p>  <p>Hybrid</p>  </div> </div> <p style="text-align: right; margin-right: 50px;">if 1 head</p> |    |
|                | (c)(ii) |  <p style="text-align: right;">shape of orbital</p> <p style="text-align: right;"><math>4\sigma - 1\pi = 3</math></p> <p style="text-align: right;">half-headed - 2</p> <p>Note:<br/>           Correct shape of orbital and overlapping – 1 mark<br/>           Label 4 sigma bonds and 1 pi bond – 1 mark<br/>           Label orbital and atoms – 1 mark</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| <b>TOTAL</b>   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>MAXIMUM</b> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |

| NO. | PART     | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | (a)      | $PV = n_{\text{Ne}}RT \quad @$ $(2)(5) = (n_{\text{Ne}})(0.08206)(25 + 273.15)$ $n_{\text{Ne}} = 0.409 \text{ mol}$ $n_{\text{H}_2} = 0.51 \text{ mol} - 0.409 \text{ mol} = 0.101 \text{ mol} \quad @$ <p style="text-align: right;">if marked Ne<br/>sahh<br/>hill<br/>hah</p> $PV = n_{\text{H}_2}RT$ $(P_{\text{H}_2})(5) = (0.101)(0.08206)(25 + 273.15) \quad \text{ecf}$ $P_{\text{H}_2} = 0.49 \text{ atm} \quad \text{unt inirt}$ <p><b><u>ALTERNATIVE SOLUTION:</u></b></p> $P_T = \frac{n_T RT}{V}$ $= \frac{(0.51 \text{ mol})(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(25 + 273.15 \text{ K})}{5.0 \text{ L}}$ $= 2.49 \text{ atm}$ $P_T = P_{\text{Ne}} + P_{\text{H}_2} \quad @ \quad \text{ecf}$ $P_{\text{H}_2} = 2.49 - 2$ $= 0.49 \text{ atm}$ |
|     | (b)(i)   | <p>CO<sub>2</sub> will undergoes <b>sublimation</b> &amp; <i>subline</i></p> <p><b>Solid CO<sub>2</sub> turn to gas</b> because <u>the triple point is above 1 atm.</u></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|     | (b)(ii)  | <p>TC line has a <b>positive gradient</b>. <i>(positive slope)</i></p> <p>Thus the <b>melting point of solid CO<sub>2</sub> increases with increasing pressure.</b> <i>(freezing pt)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|     | (b)(iii) | Supercritical fluid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|     |          | <b>TOTAL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|     |          | <b>MAXIMUM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

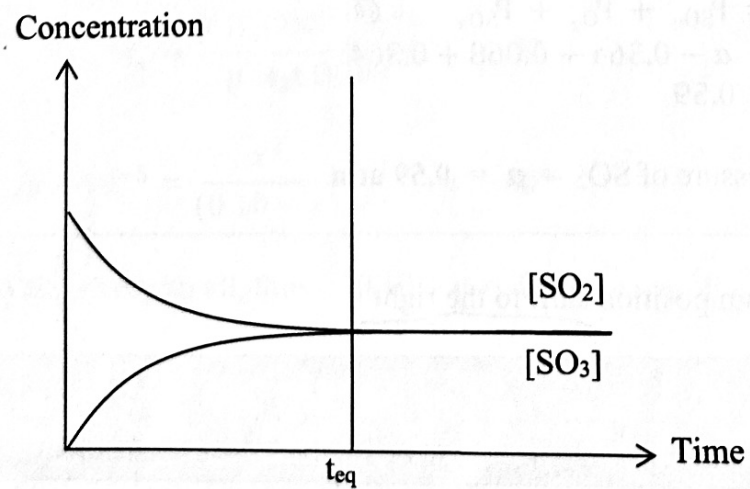
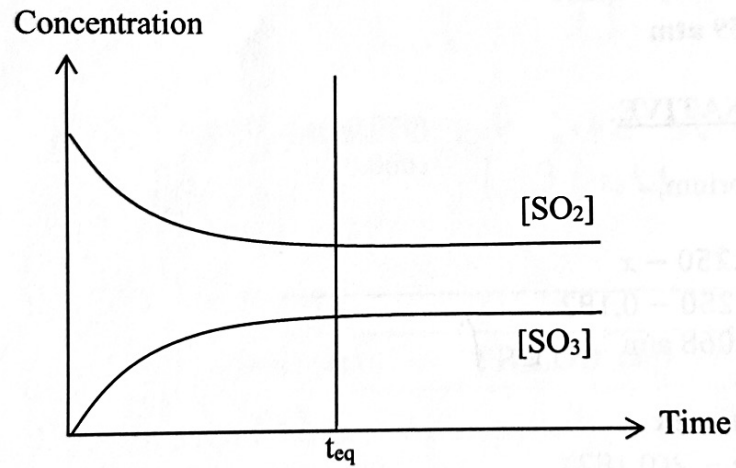
| NO. | PART | SCHEME |
|-----|------|--------|
|-----|------|--------|

5

(a)



**ALTERNATIVES:**





| NO.             | PART                          | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MA                         |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
|-----------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|----------------------------------------------------------|----------------------------|-------------|------|---|---|------------|----|----|----|-----------------|----------|---|---|--|
| 6               | (a)(i)                        | <p>When strong acid is added:</p> $\text{H}^+ + \text{NO}_2^- \rightarrow \text{HNO}_2 @$ $\text{H}_3\text{O}^+ + \text{NO}_2^- \rightarrow \text{HNO}_2 + \text{H}_2\text{O}$ <p><i>arrow long<br/>phase ut<br/>mrt.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
|                 | (a)(ii)                       | <p>New <math>[\text{HNO}_2] = \frac{0.20 \times 150}{500} @</math></p> <p><math>= 0.060 \text{ M}</math></p> <p>New <math>[\text{NO}_2^-] = \frac{0.10 \times 350}{500} @</math></p> <p><math>= 0.070 \text{ M}</math></p> <p><math>\text{pH} = \text{pK}_a + \log \frac{[\text{NO}_2^-]}{[\text{HNO}_2]}</math> <i>check ini dulu</i></p> <p><math>= -\log 4.95 \times 10^{-4} + \log \frac{(0.070)}{(0.060)}</math> <i>cek jika selk ada<br/>[ ] selk</i></p> <p><math>= 3.37</math></p> <div style="text-align: right;"> <math display="block">\begin{array}{c c c} m4 &amp; m2 &amp; m3 \\ \hline \checkmark &amp; [1/(1)] &amp; [1/(1)] \checkmark 3m \\ \hline \times &amp; (1) &amp; (1) 1m \\ \hline \times &amp; [ ] &amp; [ ] 2m \end{array}</math> </div>                                                                           |                            |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
|                 | (b)(i)                        | <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th></th><th><math>\text{CH}_3\text{COOH (aq)}</math></th><th><math>\rightleftharpoons \text{CH}_3\text{COO}^- \text{(aq)}</math></th><th><math>+ \text{H}^+ \text{(aq)}</math></th></tr> </thead> <tbody> <tr> <td>[ ] initial</td><td>0.10</td><td>0</td><td>0</td></tr> <tr> <td>[ ] change</td><td>-x</td><td>+x</td><td>+x</td></tr> <tr> <td>[ ] equilibrium</td><td>0.10 - x</td><td>x</td><td>x</td></tr> </tbody> </table> $K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} @$ $1.8 \times 10^{-5} = \frac{x^2}{(0.10 - x)} @$ <p>(Since x is very small, thus <math>0.10 - x = 0.10</math>) <i>asat ni keane<br/>sela</i></p> $1.8 \times 10^{-5} = \frac{x^2}{(0.10)}$ $x = 1.3416 \times 10^{-3}$ |                            | $\text{CH}_3\text{COOH (aq)}$ | $\rightleftharpoons \text{CH}_3\text{COO}^- \text{(aq)}$ | $+ \text{H}^+ \text{(aq)}$ | [ ] initial | 0.10 | 0 | 0 | [ ] change | -x | +x | +x | [ ] equilibrium | 0.10 - x | x | x |  |
|                 | $\text{CH}_3\text{COOH (aq)}$ | $\rightleftharpoons \text{CH}_3\text{COO}^- \text{(aq)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $+ \text{H}^+ \text{(aq)}$ |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
| [ ] initial     | 0.10                          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0                          |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
| [ ] change      | -x                            | +x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | +x                         |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |
| [ ] equilibrium | 0.10 - x                      | x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | x                          |                               |                                                          |                            |             |      |   |   |            |    |    |    |                 |          |   |   |  |

| NO.                                                                                       | PART                   | SCHEME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                           |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |
|-------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--|--|--|----------------------|-------------------|-------|---|---------------------|-------|------|------|----------------------|------------------------|-------------|------|
|                                                                                           | (b)                    | <table border="1"> <tr> <td colspan="4"><math>2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})</math></td> </tr> <tr> <td><math>P_{\text{initial}}</math></td> <td><math>P_{\text{SO}_2}</math></td> <td>0.250</td> <td>0</td> </tr> <tr> <td><math>P_{\text{change}}</math></td> <td><math>-2x</math></td> <td><math>-x</math></td> <td><math>2x</math></td> </tr> <tr> <td><math>P_{\text{equilb.}}</math></td> <td><math>P_{\text{SO}_2} - 2x</math></td> <td><math>0.250 - x</math></td> <td><math>2x</math></td> </tr> </table> <p> <math>2x = 0.364</math> @<br/> <math>x = 0.182</math> </p> <p> <math>P_{\text{O}_2} = 0.250 - 0.182</math> @<br/> <math>= 0.068 \text{ atm}</math> <i>ecf</i> </p> <p>           Partial pressure of <math>\text{SO}_2</math> at equilibrium:<br/> <math>P_{\text{SO}_2} = 0.658 - 0.364 - 0.068</math> @<br/> <math>= 0.226 \text{ atm}</math> </p> <p>           Initial pressure of <math>\text{SO}_2</math>:<br/> <math>P_{\text{SO}_2} - 2(0.182) = 0.226</math><br/> <math>P_{\text{SO}_2} = \mathbf{0.59 \text{ atm}}</math> </p> <p><b><u>ALTERNATIVE:</u></b></p> <p>At equilibrium,</p> <p> <math>P_{\text{O}_2} = 0.250 - x</math><br/> <math>= 0.250 - 0.182</math><br/> <math>= 0.068 \text{ atm}</math> <i>ecf</i> </p> <p> <math>P_{\text{SO}_2} = a - 2x</math><br/> <math>= a - 2(0.182)</math><br/> <math>= (a - 0.364) \text{ atm}</math> </p> <p> <math>P_{\text{T}} = P_{\text{SO}_2} + P_{\text{O}_2} + P_{\text{SO}_3}</math> @<br/> <math>0.658 = a - 0.364 + 0.068 + 0.364</math><br/> <math>a = 0.59</math> </p> <p>Initial pressure of <math>\text{SO}_2 = a = \mathbf{0.59 \text{ atm}}</math></p> | $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ |  |  |  | $P_{\text{initial}}$ | $P_{\text{SO}_2}$ | 0.250 | 0 | $P_{\text{change}}$ | $-2x$ | $-x$ | $2x$ | $P_{\text{equilb.}}$ | $P_{\text{SO}_2} - 2x$ | $0.250 - x$ | $2x$ |
| $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                           |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |
| $P_{\text{initial}}$                                                                      | $P_{\text{SO}_2}$      | 0.250                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                                                                                         |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |
| $P_{\text{change}}$                                                                       | $-2x$                  | $-x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $2x$                                                                                      |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |
| $P_{\text{equilb.}}$                                                                      | $P_{\text{SO}_2} - 2x$ | $0.250 - x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $2x$                                                                                      |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |
|                                                                                           | (c)                    | Equilibrium position <u>shift to the right</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                           |  |  |  |                      |                   |       |   |                     |       |      |      |                      |                        |             |      |

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| NO. | PART    | SCHEME                                                                                                                                                                                                                                                                                                                                                    |
|-----|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |         | $[H^+] = x = 1.3416 \times 10^{-3} \text{ M}$ $\text{pH} = -\log [H^+] \quad @$ $= -\log (1.3416 \times 10^{-3})$ $= 2.87$ <p><b><u>ALTERNATIVE:</u></b></p> $[H_3O^+] = \sqrt{K_a \times C}$ $= \sqrt{1.80 \times 10^{-5} \times 0.100}$ $= 1.34 \times 10^{-3} \text{ M}$ $\text{pH} = -\log [H_3O^+] \quad @$ $= -\log (1.34 \times 10^{-3})$ $= 2.87$ |
|     | (b)(ii) | <p>selah curve = 0 mark</p> <p>Equilibrium point</p> <p>7</p> <p>Buffer region</p> <p>Vol. CH<sub>3</sub>COOH<br/>in ml</p> <p>M 14<br/>15<br/>at</p>                                                                                                                                                                                                     |
|     |         | TOTAL                                                                                                                                                                                                                                                                                                                                                     |
|     |         | MAXIMUM                                                                                                                                                                                                                                                                                                                                                   |